Table. 1 Comparison of chordates and non-chordates

Chordates	Non-chordates
Notochord is present	Notochord is absent
Dorsal, hollow and single nerve cord	Double ventral solid nerve cord
Pharynx perforated by gill slits	Gill slits absent
Heart is ventrally placed	Heart is dorsal or laterally placed or absent
A post anal tail is present	A post anal tail is absent
Alimentary canal is placed ventral to the nerve cord	Alimentary canal is placed dorsal to the nerve cord

chordate. In aquatic forms, pharyngeal gill slits are vascular, lamellar and form the gills for respiration. In terrestrial chordates, traces of non-functional gill clefts appear during embryonic developmental stages and disappear later. Besides the above said features, chordates are bilaterally symmetrical, triploblastic, coelomates with organ system level of organisation; they possess post anal tail, closed circulatory system with a ventral myogenic heart except in *Amphioxus*.

List the three features common to all chordates at sometime in their life.

2.4.1 Subphylum: Urochordata or Tunicata

(G. *Oura* – A tail; L. *Chord* – cord)

They are exclusively marine and are commonly called **sea squirts**. Mostly sessile, some pelagic or free swimming, exist as solitary and colonial forms. Body is unsegmented and covered by a test or tunic. Adult forms are sac like. Coelom is absent, but has an atrial cavity surrounding the pharynx. Notochord is present only in the tail region of the larval stage, hence named urochordata. Alimentary canal is complete and circulatory system is of open type. The heart is ventral and tubular. Respiration is through gill slits and clefts. Dorsal tubular nerve cord is present only in the larval stage and a single dorsal ganglion is present in the adults. Mostly hermaphrodites, development indirect and includes a free swimming **tadpole larva** with chordate characters. **Retrogressive metamorphosis** is seen (Figure 2.20).

Examples: *Ascidia, Salpa, Doliolum*

2.4.2 Subphylum: Cephalochordata

(L. *Cephalo-* 'head' ; G. *chorda* 'cord'.)

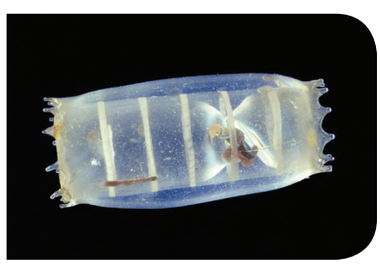
Cephalochordates are marine forms, found in shallow waters, leading a burrowing mode of life. They are small



Ascidia



Salpa



Doliolum

Figure 2.20 Examples of Urochordata

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